Snapshot of October 29, 2026. This dossier describes the four repositories as they stood that day; the projects have moved on since, and the repositories are the authority. Underlined links lead to the sources, frozen at their state of that day.

DOSSIER 1 · THE DESKTOP

gnosix

A desktop environment for Linux, written in Rust, that draws everything you see itself: the bar, the deck and its AppStack, the windows, the settings, the lock screen.

EXPERIMENTAL | 26.11-RC1 « ARDOISE » | MIT LICENCE

8

surfaces **proven** in the author's real session

9

surfaces **qualified** in an isolated session, awaiting review

2

surfaces **verified in source** only

1

engine still **experimental**: the background renderer

In one picture

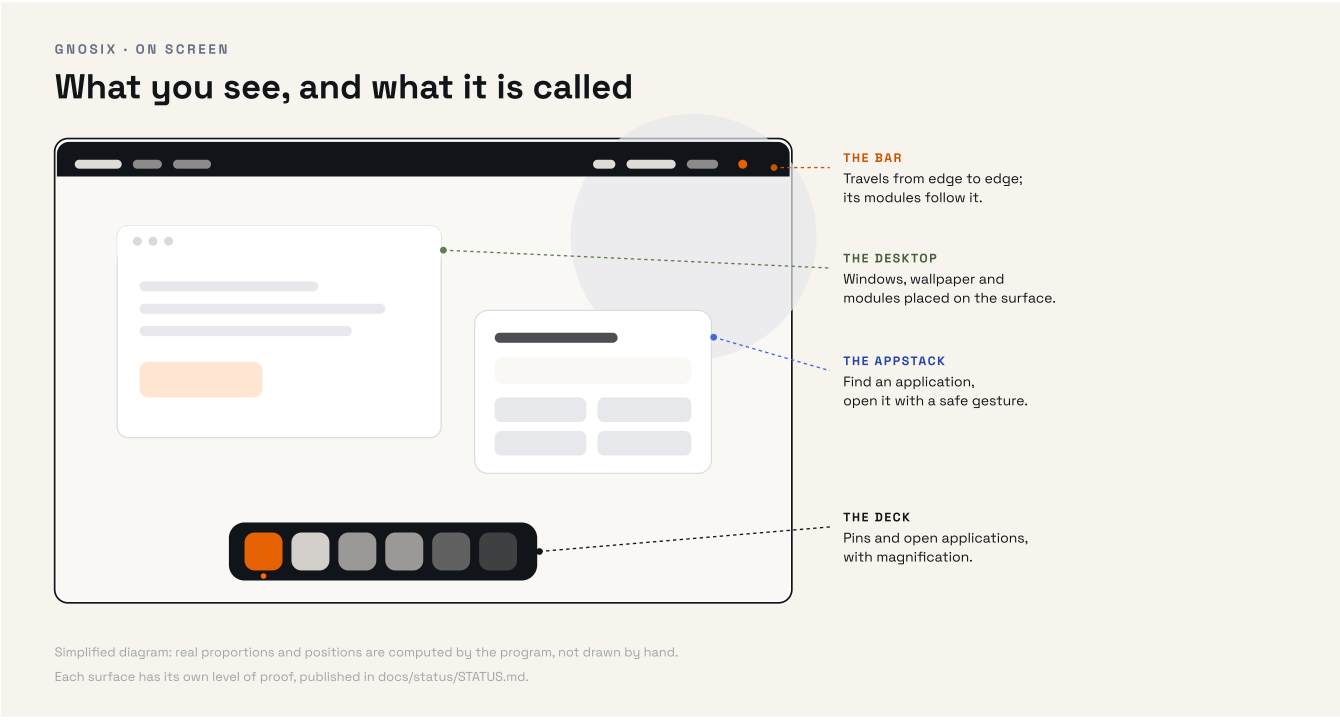
THE ANALOGY

Think of a tidy workshop. There is the **workbench** where you act, the **drawers** where you keep what must survive the night, the **shelf** where the tools sit, and the **rulebook** that says who may open what.

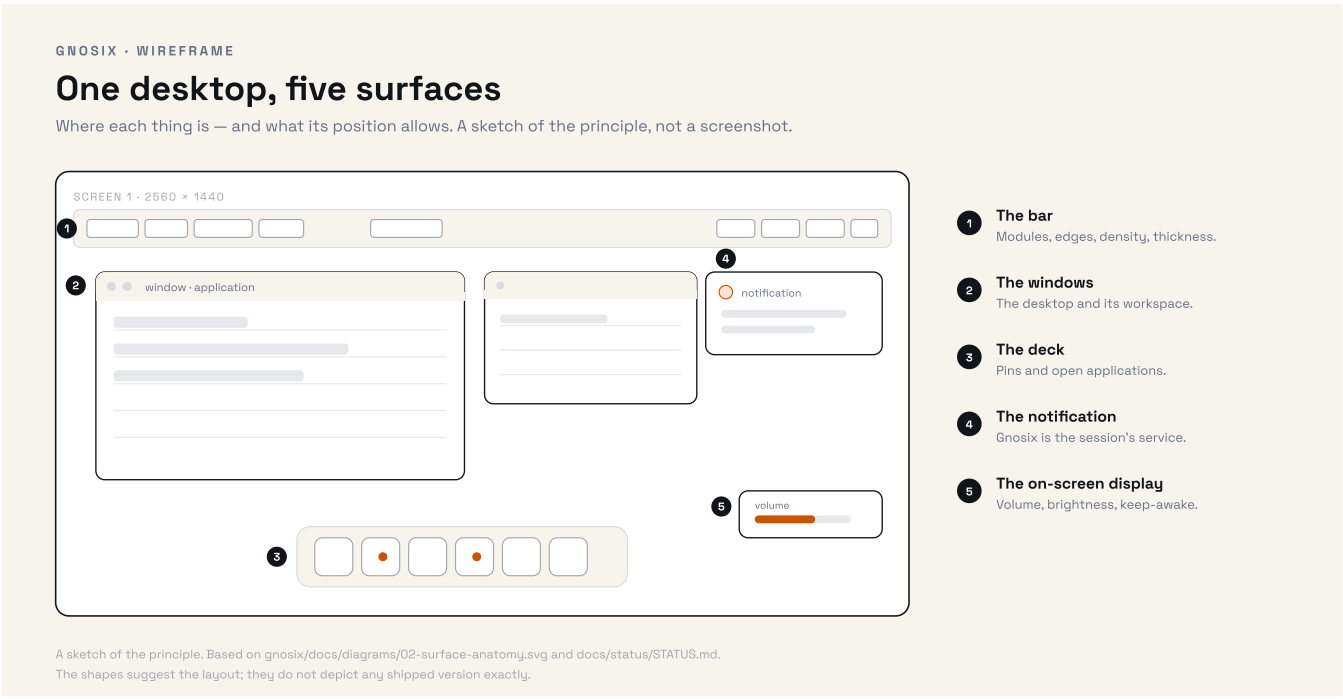
A computer desktop is exactly that. Most desktops melt all four into a single program. gnosix keeps them apart, and names each one.

What it does, concretely

Once started, gnosix replaces the session's graphical shell. It is not a decoration laid over an existing desktop: it is the desktop itself, drawn in Rust and displayed natively.



The four main surfaces. The bar can travel from one edge to another; the deck gathers the applications; the AppStack, housed in the deck, is for searching and opening; the desktop carries the windows and modules.



A sketch of the principle. The numbered markers (1 to 5) name the surfaces: bar, windows, deck, notification, on-screen display. It is a proposed layout, not a screenshot — the shapes suggest, they do not reproduce.

The bar

Persistent, on any edge of any screen. It moves along the frame; its modules follow it and turn with it. They can be reordered with the mouse. Edge, density, thickness and visibility are remembered.

The deck and the AppStack

A dock of application cards, with magnification. The AppStack searches the catalogue from there (recent, A–Z, categories, search) and launches through the launch authority. Pins are durable.

The utilities

Quick toggles (do not disturb, keep awake, microphone, Wi-Fi, Bluetooth), brightness (backlight and DDC/CI) and volume, with an on-screen display.

The session

Lock, suspend, log out, reboot, power off — through logind. Heavy actions are held back, blockers are shown.

The dashboard

Calendar, media controls (MPRIS) and system resources, extruded from the bar's clock.

The workspaces

A workspace manager with live window thumbnails — in memory only, never while the session is locked, with a list of excluded applications.

The notifications

gnosix *is* the session's notification service. Per-application rules, a "do not disturb" mode.

The lock screen

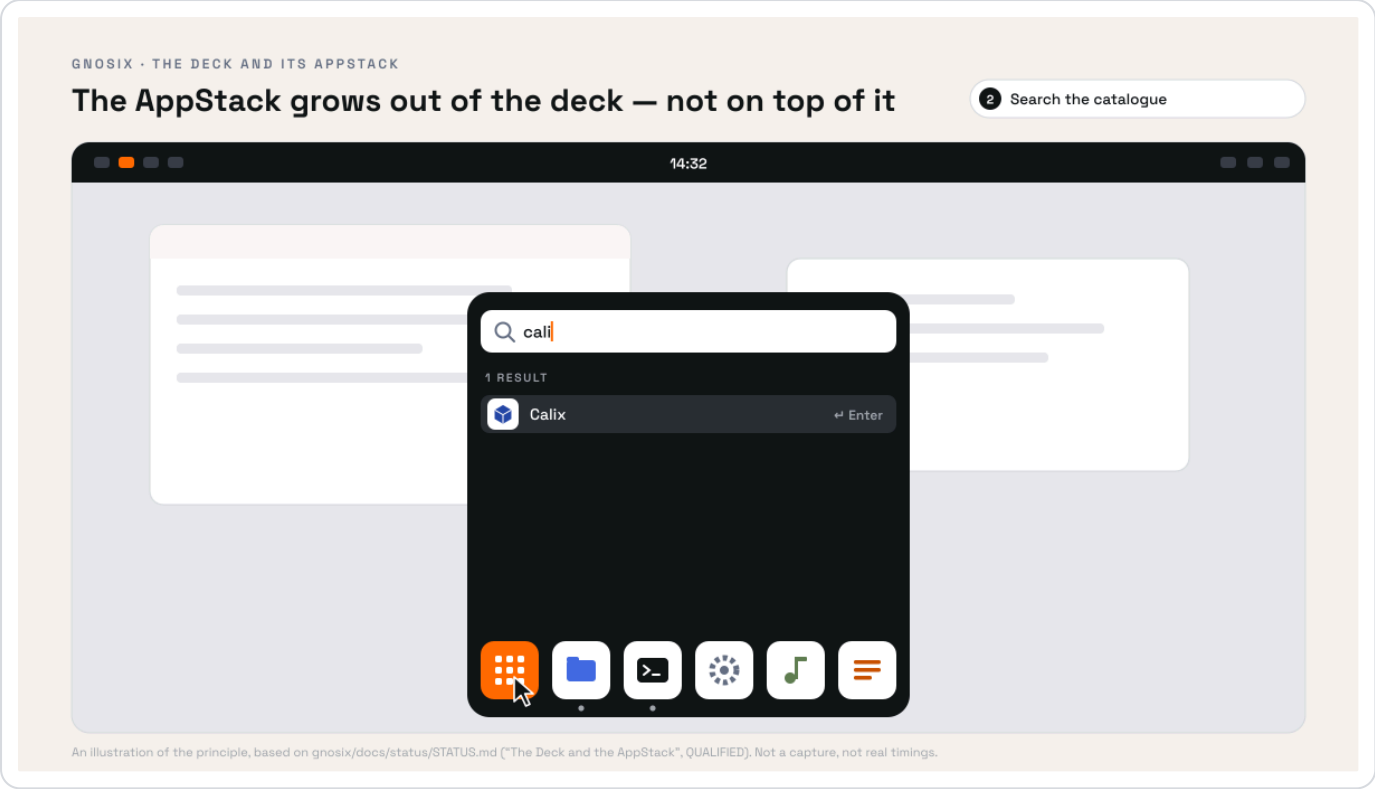
A lock screen with PAM, automatic lock after inactivity, and its settings.

The settings

A single surface: bar, deck, surfaces, displays (resolution, refresh rate, scale, rotation, position, with keep-or-revert), default applications, notifications, language, wallpaper.

Language and icons

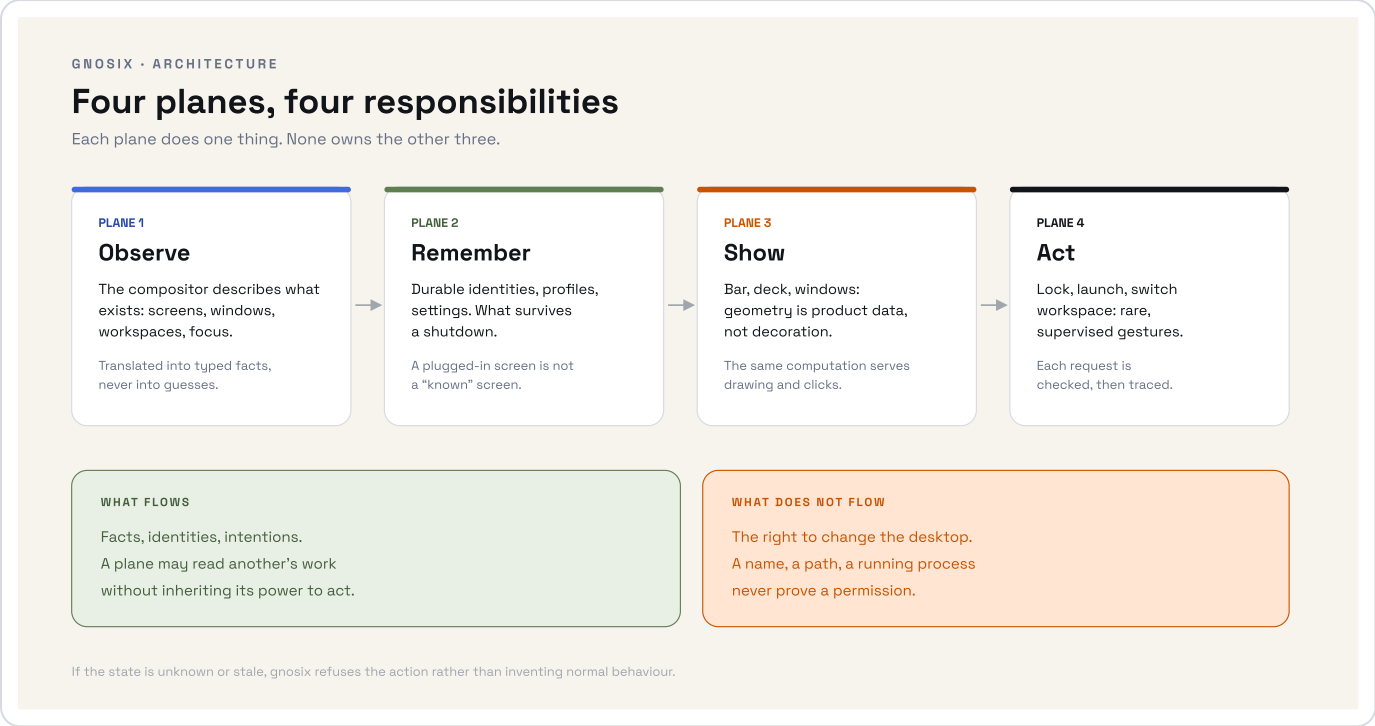
English and French throughout, chosen in the settings, with a coverage check. Tabler icons drawn natively.



The AppStack comes out of the deck. A single shape stretches out of the deck, instead of a window laid on top; the search lives inside it, then everything folds back. Launching goes through the launch authority, which grants or refuses. An illustration of the principle: neither a screen capture nor real timings.

Under the hood: four planes

It is the project's most important architectural decision, and the easiest to explain: **observing, remembering, showing and acting are four different trades.** gnosix gives them four distinct planes.



The central rule. Passing data from one plane to another never transfers the right to act. A name, a path or a running process does not prove a permission.

The observation plane translates the compositor's facts into a typed, deterministic model: screens, workspaces, windows, focus, visibility. This model decides nothing; it describes.

The durable plane keeps identities and profiles. A plugged-in screen is not automatically "the known screen": the association between the session identity and the durable identity is explicit, and can deliberately fail.

The presentation plane projects the current state onto the screen. Geometry there is product data, not decoration: the same geometry is used to paint and to decide whether a click hits its target.

Finally, the authority plane carries out the rare gestures: locking, launching an application, switching workspaces. Each request carries a typed target and a freshness boundary. If the state is unknown or stale, the system refuses rather than inventing normal behaviour.

Two ways to build a desktop

QUESTION	"MONOLITHIC" DESKTOP	SEPARATED DESKTOP (GNOSIX)
Who knows the state?	The main program, a bit everywhere	A single, explicit model
Who may act?	Often the same program	Narrow, authenticated services
What happens if the state is unclear?	A plausible state is assumed	The action is refused
How do you prove a feature works?	"It seems to work"	A named, published level of proof

Neither approach is "the right one" in absolute terms. The second costs more work, but makes failures explainable and permissions verifiable.

What "proven" means here

The repository distinguishes five states, and publishes them surface by surface. This is not rhetoric: it is what stops a successful demo from turning into a general promise.

STATE	WHAT IT MEANS	EXAMPLE
PROVEN	Qualified in source, observed in the real session, confirmed by the author.	Notifications, French and English, fluidity.
QUALIFIED	Qualified in source and in an isolated Wayland session; deployed on the author's machine, confirmation pending.	The bar, the deck, the settings, the lock screen.
SOURCE-QUALIFIED	Contracts and invariants are exercised in source; broader real behaviour is not implied.	Durable screen identity, accessibility mechanics.
EXPERIMENTAL	Active work, limited claim.	The wallpaper rendering engine.
OPEN	A named obligation remains unresolved.	Full product acceptance.

What it changes, depending on who you are

Curious Linux user

You can try a desktop unlike the others, on Arch Linux with Hyprland. Set aside some time: it is not yet a turnkey install for the general public.

Developer or integrator

The repository is a rare reference on separating observation, durable state, presentation and authority. The typed contracts and invariants are documented and testable.

Decision-maker or funder

The value is not "one more desktop". It is a method: a visible product whose every claim is bounded, dated and verifiable. The risk is owned and stated.

What is still open

THE LIMITS, PLAINLY

- **Not yet a desktop for others.** 26.11-rc1 is an experimental public pre-release, used daily by its author — not a stable product.
- **Screen-reader accessibility is not claimed.** Keyboard navigation and focus exist; real use by a blind person has not been reviewed.
- **Keyboard control of the bar is designed, not built.**
- **Third-party modules are designed, not built.** A threat model has to come first.
- **The native wallpaper remains experimental.**

Going further

> TECHNICAL — LANGUAGES, BUILDING BLOCKS, LICENCE

> CHECKING FOR YOURSELF